LESSONS LEARNED

1. Introduction

This document summarizes the knowledge and practical skills developed through the Network Traffic Analysis and Security Investigation project using Wireshark.

The project is intended to improve understanding of network protocols, packet capture, network troubleshooting, and basic cybersecurity investigation.

2. Understanding Network Traffic

Network communication involves the exchange of packets between devices. Each packet contains information that helps explain how communication occurs.

Through Wireshark, an investigator can inspect packet timestamps, source and destination addresses, protocol information, and other available packet fields.

This helps transform network communication into evidence that can be examined during troubleshooting.

3. Lessons Learned About DNS

DNS translates domain names into IP addresses when resolution is required.

The dns display filter helps identify DNS queries and their responses in a packet capture.

Key lesson: A missing DNS response does not automatically indicate a DNS server failure. Possible explanations include DNS caching, an incorrect capture interface, capture timing, or network filtering.

4. Lessons Learned About ICMP

ICMP is used for network diagnostic functions, including the echo requests and replies commonly associated with ping.

The icmp filter helps distinguish these packets from other traffic.

Key lesson: Ping can help test reachability, but a failed ping does not always mean that a host is offline. Firewalls or network policies may block ICMP traffic.

5. Lessons Learned About TCP

TCP provides reliable, connection-oriented communication.

The TCP three-way handshake generally consists of:

- SYN - the client requests to establish a connection.
- SYN - ACK - the receiving endpoint acknowledges the request and responds.
- ACK - the initiating endpoint acknowledges the response.

Wireshark can help identify these packets and inspect TCP flags.

Key lesson: A TCP reset or retransmission is not automatically evidence of malicious activity. It must be interpreted in context.

6. Importance of Display Filters

Display filters help reduce the number of packets shown and make investigations easier to follow.

Examples include:

- dns for DNS traffic.
- icmp for ICMP traffic.
- tcp for TCP traffic.
- tcp.flags.syn == 1 for packets with the TCP SYN flag set.

Key lesson: Choosing an appropriate filter makes packet analysis more efficient, but filtering does not create evidence that was not captured.

7. Network Troubleshooting Skills

The project combines packet analysis with Windows diagnostic commands.

Commands such as ipconfig, ping, and nslookup provide information that can be compared with the packet capture.

Key lesson: Effective troubleshooting involves gathering evidence, checking possible causes systematically, and confirming the result after making a change.

8. Challenges and Possible Solutions

Challenge	Possible troubleshooting action
No packets appear	Confirm that the correct interface is selected and capture is running.
DNS response is missing	Repeat the lookup and inspect DNS configuration, caching, and capture timing.
Ping receives no replies	Check local connectivity and consider whether ICMP is blocked.
TCP handshake is incomplete in the capture	Verify the capture interface and inspect the connection again.
Too many packets appear	Apply a suitable display filter.

These are troubleshooting possibilities, not claims that every problem occurred during the project.

9. Security and Privacy Lessons

Packet captures can contain sensitive information about devices and network activity.

Before publishing evidence, it is important to inspect screenshots and capture files, remove sensitive information where appropriate, and ensure that the investigation was authorized.

Key lesson: Network investigation must be performed responsibly, with respect for privacy and permission boundaries.

10. Skills Developed

After completing and documenting the practical exercises, the project can demonstrate experience in:

- Wireshark packet capture and analysis.
- DNS, ICMP, and TCP fundamentals.
- Network troubleshooting using Windows tools.
- Display-filter usage.
- Technical documentation and evidence handling.
- Basic cybersecurity investigation practices.

Only claim skills that have been practiced and verified.

11. Future Improvements

Possible improvements include:

- Investigating a controlled network with multiple virtual machines.
- Comparing successful and unsuccessful DNS lookups.
- Studying TCP retransmissions using a repeatable test.
- Creating a network troubleshooting flowchart.
- Adding sanitized screenshots and verified findings.
- Publishing the completed report and project documentation on GitHub.

12. Conclusion

This project provides a foundation for further learning in network administration and cybersecurity. The most important lesson is to base conclusions on captured evidence, use diagnostic tools systematically, and document both findings and limitations clearly.